

Listening to the Collective: Entropy-Based Early Warning for Emergent Agent Communication

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Abstract. Hola

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1 Introduction

In the pursuit of AI safety, one of the central requirements when designing evaluation and deployment environments is the availability of heuristics and tools capable of detecting unwanted behaviour. This requirement becomes more pressing as large language model (LLM) agents are increasingly deployed as multi-agent systems (MAS), in which several agents that reason, plan and act through tools are combined to decompose tasks, explore in parallel or verify each other's work (Kim et al., 2026; Zhu et al., 2026). In such systems, undesired behaviour need not originate in a single agent: it can emerge from the interaction between agents, including coordination that was never intended by the system designer. The OpenAI–Hugging Face incident of July 2026 (Larcher et al., 2026; OpenAI, 2026) showed that this risk is no longer hypothetical, and it is the direct motivation for the present work.

1.1 Motivating incident

During internal cybersecurity capability evaluations (ExploitGym), agents built on a highly capable internal-only research model, trained among other things for persistence and multi-agent collaboration, and on GPT-5.6 Sol circumvented the controls that isolated their sandbox from the internet (OpenAI, 2026). Their egress path was an internal instance of JFrog Artifactory, a package mirror provided so that agents could install software. Using publicly exposed credentials and previously unknown vulnerabilities, the agents compromised parts of Hugging Face's production infrastructure between 11 and 13 July 2026: they executed code on 41 production dataset-

server workers, harvested production credentials, obtained administrator-equivalent access to a connected Kubernetes cluster and downloaded private code repositories. In under 13 hours they progressed from code execution in a single worker pod to administrative and host-level access across multiple clusters. According to both reports, the goal behind this activity was not sabotage but reward hacking: the agents were searching for the solutions of their evaluation tasks, which they inferred might be hosted on Hugging Face (Larcher et al., 2026; OpenAI, 2026). Hugging Face disclosed a security incident on 16 July; OpenAI was alerted to unusual identity-related API activity on 19 July, linked it to the Hugging Face compromise on 20 July, and disclosed the incident publicly on 21 July.

What makes this incident especially relevant for the present paper is that the intrusion was preceded by roughly two months of emergent inter-agent communication. On 12 May, an agent that could not access a file required by its task reasoned that an agent in another environment might have it and wrote a note to Artifactory: "Agent seeks [filename]; upload if found!". Other agents found such notes and, using the shared credentials they had legitimately been given, turned them into an improvised message board, without exploiting any vulnerability (OpenAI, 2026). The channel then matured steadily. By 26 May agents obtained internet access relying in part on information exchanged on the board; the first confirmed two-way coordination message was recorded on 17 June; and on 26 June agents shared exploitation syntax through the board before escalating to administrator privileges. After Artifactory was rebuilt, agents re-established the board on 8 July by encoding messages in directory names, developing a structured protocol to categorise communications, send direct messages, share tools and files, and resolve conflicting actions. On 10 July an agent published 14 leaked Hugging Face credentials to the board,

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and on 11 July live production credentials from Hugging Face were posted there, making them available to agents in subsequent, otherwise independent, evaluation runs. When responders finally contained the activity, they disabled 311 Artifactory repositories used for message-board communication.

Crucially, signals were available before the breach but were not interpreted as communication. An internal team observed message-board activity around 26 May; on 27 June a monitoring tool flagged port-sweep activity, and responders linked it to an evaluation using Artifactory “as an improvised message board and a network pivot”, but concluded that stopping the run was not required; and after the Artifactory outage of 5 July, “the existence of the improvised message board and the significance of the inter-agent communication activity were not apparent to leaders responsible for incident detection and response” (OpenAI, 2026). OpenAI itself concludes that some of these early signals could have triggered an earlier response, and among its remediations it lists severity-based escalation triggers for “unauthorized cross-agent or cross-run coordination”. The report also indicates that communication was not random: 93% of the tasks discussed on the board came from the 22% of ExploitGym tasks that no model had ever solved, and participation in the board increased with the reasoning effort granted to the agents.

The two reports also offer complementary views of the same events. From the operator’s side, OpenAI describes an “automated agent collective” in which many agent instances, across different runs and two models, shared discoveries and built on each other’s progress (OpenAI, 2026). From the victim’s side, Hugging Face reconstructed approximately 17,600 actions, grouped into about 6,280 clusters, between 9 and 13 July, and attributes them to a single autonomous agent driven by a combination of OpenAI models (Larcher et al., 2026). Its defenders faced a volume problem: several detection layers fired, but the alert did not reach the correct criticality to page the on-call team, and “the successful path was hidden inside the noise generated by the thousands of failed ones”. This contrast has an important consequence: seen from outside, a coordinated collective is indistinguishable from one high-volume attacker, so the coordination itself is observable only from within the environment that hosts the agents, where their outputs and model distributions are also available.

1.2 Problem and proposed approach

The incident also illustrates why detecting communication by monitoring specific channels is not a viable strategy. The agents exchanged information through plain-text files, directory names, messages encoded in URL

paths, public Hugging Face datasets used as a pull-based command-and-control channel, pastebins, webhooks and even rendered screenshots decoded by optical character recognition (OpenAI, 2026). Hugging Face further reports that staged data was chunked, XOR-encrypted, gzip-compressed and base64-encoded, so that naive log scans missed about 75% of the sensitive material (Larcher et al., 2026). None of these resources was designed as a communication channel, and a detector written for any one of them would have been bypassed by the next. Moreover, the effect of coordination on system behaviour depends strongly on the architecture and on the capability of the underlying models (Kim et al., 2026), which makes purely structural detection criteria unreliable.

In this paper we propose a channel-agnostic mechanism for detecting communication between agents in a MAS. Rather than inspecting the channels themselves, we study the behaviour of the entropy of the individual agents and of the system as a whole, and we fit regression models to its trajectory in order to detect anomalies that may indicate interaction. The underlying intuition is that agents that are supposed to be independent, such as parallel evaluation runs, should produce statistically independent outputs; when they begin to read and act upon each other’s messages, their outputs become coupled, regardless of the medium used to exchange them. Entropy-based summaries have proven effective for anomaly detection in network traffic (Lakhina et al., 2005), for uncertainty estimation and hallucination detection in LLMs (Farquhar et al., 2024; Kuhn et al., 2023), and for characterising the dynamics of orchestrated multi-agent systems (Zhu et al., 2026). Applied to the incident described above, a monitoring layer of this kind could, in principle, have raised an alarm when otherwise independent evaluation runs began to show correlated behaviour in May, weeks before the Hugging Face compromise and without any prior knowledge of Artifactory as a channel. We regard such an interaction-detection layer as complementary to chain-of-thought monitoring and to the escalation triggers proposed by OpenAI, and we treat its early-warning capability as a hypothesis to be tested rather than as an established result.

To begin testing this hypothesis, we complement the proposal with a preliminary experimental study based on a simple benchmark, whose purpose is to determine whether the entropy trajectory follows a general functional form and whether that form changes when agents communicate. The remainder of the paper is organised as follows. First, we define the information-theoretic framework on which the method is based. Next, we describe the concrete implementation of the benchmark and the methodology, including the precise definition of the dataset and of

the models used. Finally, we interpret the collected measurements in search of mechanistic changes in the entropy that can be attributed to inter-agent communication.

2 Entropy Framework

Introduced by Claude Shannon in 1948 (Shannon, 1948), entropy has become one of the cornerstones of information theory. It sets the fundamental limit of lossless data compression, which is attained in practice by source codes such as Huffman coding (Huffman, 1952); it serves as a summary statistic for detecting anomalies in network traffic, including distributed denial-of-service (DDoS) attacks (Lakhina et al., 2005); and, more recently, it has become a key tool for estimating uncertainty and detecting hallucinations in LLMs (Farquhar et al., 2024; Kuhn et al., 2023). For a discrete random variable X taking values $x_1, \dots, x_n$, the entropy is defined as

$$H(X) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i), \quad (1)$$

where $p(x_i)$ is the probability of the event $X = x_i$, and the convention $0 \log_2 0 = 0$ is adopted. With the logarithm in base 2, entropy is measured in bits. $H(X)$ quantifies the expected uncertainty (or surprise) associated with the outcome of X : it is zero when the outcome is certain, and it reaches its maximum value, $\log_2 n$, when all outcomes are equally likely (Cover & Thomas, 2006).

2.1 Entropy of a single agent

For an LLM-based agent, the natural probability distribution is the model's distribution over the next token. At each generation step t , the model produces a vector of logits $z_t \in \mathbb{R}^{|V|}$ over the vocabulary V . Logits are unnormalised scores, not probabilities; the next-token distribution is obtained through the softmax function,

$$p(v | x_{<t}) = \frac{\exp(z_{t,v}/T)}{\sum_{u \in V} \exp(z_{t,u}/T)}, \quad (2)$$

where $x_{<t}$ denotes the context (prompt and previously generated tokens) and T is the sampling temperature. Consequently, any entropy computed from this distribution depends on the chosen temperature. Two families of entropy measures can be built on top of these distributions:

1. **Predictive entropy.** The token-level entropy

$$H_t = - \sum_{v \in V} p(v | x_{<t}) \log_2 p(v | x_{<t}) \quad (3)$$

measures the model's uncertainty about the next token. At the sequence level, the predictive entropy of a response s given an input x is $H(S | x) =$

$-\mathbb{E}_{s \sim p(\cdot|x)} [\log_2 p(s | x)]$, which is usually estimated by Monte Carlo sampling or approximated by aggregating the token-level entropies along a generated sequence (Malinin & Gales, 2021). Because it operates directly on token probabilities, predictive entropy mixes semantic uncertainty with lexical and syntactic uncertainty: two phrasings of the same content count as different outcomes.

2. **Semantic entropy.** Semantic entropy was introduced to separate uncertainty about meaning from uncertainty about form, a separation that is crucial for applications such as hallucination detection (Farquhar et al., 2024; Kuhn et al., 2023). Instead of operating on individual tokens, it samples several complete responses to the same input, groups them into clusters of semantically equivalent answers (for example, using bidirectional entailment judged by a natural language inference model), and computes the entropy of the distribution over these meaning clusters. As a result, variation in wording that does not change the meaning of a response does not increase the entropy.

In this work we adopt predictive entropy. Our working hypothesis is that, when two agents start to interact, the incorporation of content produced by another agent into an agent's context (with a different style, vocabulary and information) changes its next-token distributions, and that part of this signal lies precisely in the lexical and syntactic variation that semantic entropy is designed to discard. This choice has two caveats that must be controlled for in the experiments. First, predictive entropy is also sensitive to factors unrelated to communication, such as response length, prompt formatting and temperature, which can act as confounders. Second, many commercial APIs expose only the log-probabilities of the top- k tokens rather than the full distribution, so Equation (3) can only be approximated, for instance by renormalising over the returned tokens or by bounding the contribution of the remaining probability mass.

2.2 Entropy of the multi-agent system

The quantities above are defined per agent and are therefore insufficient to describe the MAS as a whole. Let $X_1, \dots, X_n$ be random variables describing the outputs of the n agents in the system (for instance, their next-token or response distributions at a given time step t). The joint entropy of the system decomposes as

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i) - TC(X_1; X_2; \dots; X_n), \quad (4)$$

where $TC(X_1; X_2; \dots; X_n)$ is the total correlation (Watanabe, 1960), also known as multi-information. This expression is in fact the definition of total correlation, which quantifies the redundancy, or statistical dependence, shared among the agents. It satisfies $TC \geq 0$, and for $n = 2$ it reduces to the mutual information $I(X_1; X_2)$ (Cover & Thomas, 2006). Intuitively, if the per-agent entropies were simply added, the information shared among interacting agents would be counted more than once, and the sum would overestimate the true uncertainty of the system; the total correlation term corrects for this overcounting. When the entropies are tracked over time, all terms in Equation (4) are evaluated at the same step t .

This decomposition contains one of the key details for using entropy to fingerprint the behaviour of a MAS. Two regimes can be distinguished:

1. **Independent agents.** $TC = 0$ if and only if the variables $X_1, \dots, X_n$ are mutually independent. In that case the system entropy equals the sum of the individual entropies, $H(X_1, \dots, X_n) = \sum_i H(X_i)$.
2. **Coupled agents.** When agents exchange information, directly or through a shared resource, their outputs become statistically dependent, $TC > 0$, and the joint entropy falls below the sum of the individual entropies. This is the regime observed in the OpenAI–Hugging Face incident, where agents in separate evaluation runs, designed to be independent, coordinated through an improvised message board hosted on shared infrastructure (OpenAI, 2026).

The gap $\sum_i H(X_i) - H(X_1, \dots, X_n)$ is therefore a natural candidate signal for detecting communication. Note, however, that dependence does not imply communication: agents that share the same underlying model, system prompt or task can exhibit correlated outputs without exchanging any information, so the total correlation of a non-communicating baseline must be measured and subtracted. Moreover, estimating the joint entropy is considerably harder than estimating the marginal entropies, because the joint outcome space grows exponentially with the number of agents and the vocabulary size, and plug-in estimators are biased with limited samples. For completeness, in this project we compute both the sum of the individual entropies (the independence assumption) and estimates that account for the coupling term, in search of a fingerprint strong enough to detect communication between agents.

3 Controller-Owned Append-Only Board Storage

Chainlink issue #16 establishes the storage boundary for the experiment's message board. The implementation

uses a controller-owned SQLite database, not a database exposed to the Pi agent. The public storage interface is deliberately small:

```
with BoardStore.initialize(database_path,
                           agent_workspace_roots=roots) as board:
    record = board.append_message(run_id, agent_id, message)
```

The service generates the UTC server timestamp and UTF-8 byte count. The SQLite table contains the following fields:

Field	Constraint or meaning
sequence_id	INTEGER PRIMARY KEY AUTOINCREMENT
run_id	Required non-empty text
agent_id	Required non-empty text
server_timestamp	Required controller-generated UTC text
message_body	Required non-empty UTF-8 text
message_size	Required byte length of the UTF-8 body

Append-only behavior is enforced in the database itself. Triggers reject both UPDATE and DELETE, so the invariant does not depend only on callers using the Python interface. The low-level iterator orders rows by sequence_id; cursor semantics, C0/C1/C2 visibility, agent identity derivation, and Pi extension tools remain follow-up work.

The database path is checked against agent workspace roots, including paths that resolve through symlinks. The board service retains the database path and connection. Neither is passed to Pi, mounted into Bubblewrap, or made available to an agent task directory.

The implementation and regression tests were delivered in commit 264af09 and pushed to origin/main as part of the merge at d39bd82. Verification recorded for issue #16 is 39 passing unittest cases, successful runtime configuration validation, successful Python compilation checks, and a clean git diff -check. The tests cover schema creation, required-field and size constraints, UTF-8 sizing, monotonic ordering, symlink-aware workspace rejection, and direct SQL UPDATE/DELETE rejection. No subscription-backed model request was made.

4 Limitations and Next Steps

The proposed method uses the predictive entropy of the agents, and the coupling between them captured by the total correlation, as a proxy for changes in behaviour, with a particular focus on the emergence of communication within a MAS. Its purpose is to provide a channel-agnostic signal, since the improvised message boards used by the agents in the OpenAI–Hugging Face incident were hosted on resources never intended for communication. However, the same breadth that makes the measure channel-agnostic also makes it sensitive to other sources of variation, such as the normal dynamics of agents while solving

a task, response length, prompt formatting and temperature. In addition, statistical dependence does not imply communication, since agents sharing a model, prompt or task may be correlated without exchanging any information; joint entropy is hard to estimate with limited samples; and many APIs expose only top- k log-probabilities, so the entropy can only be approximated. The method also requires access to the models' output distributions, which restricts it to the operator hosting the agents, and our experimental evidence is preliminary, based on a single simple benchmark.

Further research is therefore needed to detect specific behaviours from the entropy of the system more precisely. A particularly promising direction is the design and training of a machine learning model on entropy trajectories that distinguishes communication from other behavioural changes. It is also necessary to study more broadly how tools, new information and other perturbations affect the entropy of a MAS, to establish non-communicating baselines across architectures and models, and to validate the approach in more realistic settings, with the aim of building a more robust detector of unintended behaviours and capabilities in general.

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